



Corticosterone levels, leukocyte profiles, and foraging and diving behaviours of Humboldt penguins during chick rearing in Northern Chile

Miriam Lerma ^a, Camila P. Villavicencio ^b, Nicolas Luna ^c, Matías Portflitt-Toro ^c, Juan Serratosa ^{c,d},
Guillermo Luna-Jorquera ^c, Stefan Garthe ^a and Rene Quispe ^{e,f}

^aResearch and Technology Centre (FTZ), University of Kiel, Büsum, Germany; ^bInstituto de Ecología y Biodiversidad, Departamento de Ciencias Ecológicas, Universidad de Chile, Ñuñoa, Santiago, Chile; ^cCenter for Ecology and Sustainable Management of Oceanic Islands (ESMOI), Facultad de Ciencias del Mar, Universidad Católica del Norte, Coquimbo, Chile; ^dBirdLife International, David Attenborough Building, Pembroke Street, Cambridge CB3 3QZ, UK; ^eDepartamento de Biología Marina, Facultad de Ciencias del Mar, Universidad Católica del Norte, Coquimbo, Chile; ^fDepartamento de Ciencias Biológicas Animales, Facultad de Ciencias Veterinarias y Pecuarias, Universidad de Chile, Santiago, Chile

ABSTRACT

Understanding the physiology of stress in wild animals is essential for the conservation of species subject to anthropogenic perturbations. Humboldt penguins (*Spheniscus humboldti*) are exposed to increasing anthropogenic impacts in their natural habitat. In this species, females are typically smaller and dive less deep than males. In related species, the more limited foraging habitat of females and their reduced resiliency due to their smaller size were associated with higher mortality. We hypothesise that potential sex-specific differences in the foraging behaviour of Humboldt penguins may also relate to differences in their physiological stress. Here, we studied sex-specific foraging and diving behaviours and variations in plasma corticosterone levels and leukocyte profiles of Humboldt penguins during the chick-rearing period in Northern Chile. We report no evidence of sex-related differences in most foraging parameters, except that males dived significantly deeper than females. We found that plasma corticosterone levels and leukocyte profiles showed no significant differences between the sexes. Furthermore, there was no clear relationship between an individual's foraging behaviour and its plasma corticosterone level or leukocyte profile. In summary, we found no support for sex-related differences in physiological stress levels of Humboldt penguins, and no link between their foraging behaviour and their physiological stress. However, we acknowledge that our sample size is small and that more studies are needed. This study contributes with information on the physiological stress and foraging behaviours of Humboldt penguins in Northern Chile. This information can help to understand context-dependent differences in physiological parameters and foraging behaviours for the species.

ARTICLE HISTORY

Received 8 June 2021

Accepted 23 November 2022

KEYWORDS

penguin; Chile; diving; foraging; sex-related differences

Introduction

Numerous factors, including breeding, food shortages, and environmental and anthropogenic perturbations can affect physiological stress parameters in animals (Ellenberg *et al.* 2006; Kitaysky *et al.* 2010; Dantzer *et al.* 2014; Palacios *et al.* 2018). Information on physiological stress levels in wild animals can help to explain variations in a species' breeding success and changes in their behaviours (Ellenberg *et al.* 2006, 2007). For example, Wandering albatrosses (*Diomedea exulans*) and Adélie penguins (*Pygoscelis adeliae*) can adjust their foraging efforts in relation to changes in their physiological stress levels (Angelier *et al.* 2007, 2008), to balance their body condition and physiological state.

Physiological stress levels in free-living birds can be assessed by various methods, including measuring plasma corticosterone levels and leukocyte profiles.

Corticosterone is the primary glucocorticoid hormone in birds, and increased corticosterone levels in the blood form part of the body's coping mechanism, to facilitate behavioural and physiological changes to minimise the effect of stress on individuals (Landys *et al.* 2006; Angelier *et al.* 2008). Baseline corticosterone levels are involved in modulating the energy balance and regulating energy intake, storage, and mobilisation (Kitaysky *et al.* 2007, 2010). For instance, birds respond to food restriction by increasing their plasma corticosterone levels and regulating their feeding activity (Kitaysky *et al.* 2010). In contrast, energy intake may reduce circulating corticosterone levels (Angelier *et al.* 2007). Haematological parameters, such as leukocyte profiles, can also be used to assess physiological responses to stressors (Davis *et al.* 2008). Stressors typically depress the number of lymphocytes in the peripheral circulation

CONTACT Miriam Lerma  lerma@ftz-west.uni-kiel.de; Rene Quispe  rquispe@uchile.cl

 Supplemental data for this article can be accessed at <https://doi.org/10.1080/01584197.2022.2153699>

© 2022 BirdLife Australia

and induce an influx of heterophils (Davis *et al.* 2008). The ratio of heterophils to lymphocytes (H/L ratio) is thus expected to respond to challenging situations, with higher ratios associated with stronger stress responses (Davis *et al.* 2008).

Humboldt penguins (*Spheniscus humboldti*) are distributed along the Pacific Coast of South America, from Peru to Chile (De La Puente *et al.* 2013). The species has experienced substantial population decreases in recent decades, partially because it breeds in regions with increasing human presence and is thus exposed to increasing anthropogenic impacts (De La Puente *et al.* 2013; Quispe *et al.* 2020). Tilgo Island in the northern part of Chile supports a population of *ca.* 2000 Humboldt penguins, with possible overlap between foraging penguins and human activities (Quispe *et al.* 2020). In Humboldt penguins, both parents participate in chick rearing, but males are heavier and larger than females (De La Puente *et al.* 2013). In the related and also sexually dimorphic African Penguin (*Spheniscus demersus*), females were considered to have higher mortality compared to males because of a limited resilience due to their smaller size (Pichegru and Parsons 2014). In addition, in the Magellanic Penguin (*Spheniscus magellanicus*), haematological parameters differed between females and males, with females thought to be in poorer condition than males due to their different breeding roles and foraging behaviours (Moreno *et al.* 2002; Gallo *et al.* 2019). In Humboldt penguins, most foraging parameters show no sex-related differences but males dive on average deeper than females (Taylor *et al.* 2002; Raya Rey *et al.* 2013). Diving depth is an important component of the foraging behaviour of diving animals (Shealer 2001; Wearmouth and Sims 2008), and as a consequence of a more restricted foraging habitat in a vertical spatial extent and their limited resiliency (Yamamoto *et al.* 2019), females might express higher levels of stress. Although the relation between foraging behaviour and stress levels has been seldom explored in these penguins, such information is needed for understanding differences in their responses to natural or anthropogenic perturbations.

In this study, we hypothesise that Humboldt Penguin females have higher H/L ratios and higher plasma corticosterone levels than males based on that females might be more restricted in their foraging habitat to a vertical extent, in a poorer body condition than males, and with a limited resiliency (Moreno *et al.* 2002; Pichegru and Parsons 2014; Gallo *et al.* 2019; Yamamoto *et al.* 2019). We also expect sex-related differences in the foraging behaviour of Humboldt penguins that could help to explain any differences in haematological parameters. Finally, based on the

reported relationships between foraging behaviours and corticosterone levels in other species, such as Adélie penguins (Angelier *et al.* 2008), we hypothesise that there are relationships between plasma corticosterone levels before and after foraging trips, the H/L ratio, and foraging and diving behaviours in this species.

Methods

This study was carried out on Tilgo Island (29.543° S, 71.336° W) between 29 November and 5 December 2018, when Humboldt penguins were rearing one or two chicks (chicks weight: 1.1–1.6 kg). Individuals were caught opportunistically at their nest by hand and restrained in the researcher's lap, with the bird's head covered with an opaque cotton bag. A blood sample (maximum 0.5 ml) was obtained from the interdigital (foot webbing) veins using a 21-gauge needle and collected with capillaries within 3 min after capture. The captured birds were equipped with two devices: a time-depth recorder (TDR; model G5+; Cefas Technology, UK) and a GPS CatLog-S device sealed in a heat-shrink epoxy casing (3.7 × 2.2 × 0.8 cm; Catnip Technologies, Hong Kong). The TDRs were programmed to record data every 2 s, and the GPS devices were programmed to record time, latitude, and longitude every 4 min. The devices were attached to the back of the bird using water-resistant tape (TESA® tape) (for details on the attachment, see Wilson *et al.* 1997). The total weight of both devices was 22–30 g and the penguins weighed 3.6–5.2 kg, the devices thus represented <1% of their total weight.

After releasing the bird, the blood samples were separated to determine the leukocyte profiles and plasma corticosterone levels, and for molecular sex determination. Leukocyte profiles were evaluated in two blood smears from each penguin. The drop of blood was smeared and air-dried on individual glass slides and preserved for processing. Back at the base-camp, the remainder of the blood sample was centrifuged for 10 min to separate the red cells and plasma, and stored separately in liquid nitrogen. Once in the laboratory, the samples were stored at –80°C. A small amount of blood was collected on FTA cards (Whatman paper; GE Healthcare Life Sciences, Piscataway, NJ, USA) and stored at room temperature for sex determination.

Individuals were recaptured to recover the devices as soon as they returned to their nests. A blood sample was obtained from each individual immediately after capture (within 3 min) for measurement of post-trip plasma corticosterone levels, as described above. Out of a total of 18 adult penguins initially captured, 16

were recaptured and bled, and 11 GPS and eight TDR devices were recovered with data. The remaining devices fell off during the foraging trips and were not attached to the penguins when they returned, malfunctioned, or the individual was not recaptured. For example, some individuals were re-sighted without the devices upon the last day of the fieldwork and were not intended to be recaptured due to time limitations. Although we did not directly assess the effects of the devices on the birds, individuals sampled during this study continued chick rearing after the removal of the devices. Moreover, bio-logging studies had been carried out on similar penguins species and it has been suggested that no strong effects from devices are to be expected (Petersen *et al.* 2006; Raya Rey *et al.* 2013).

Once in the laboratory, blood smears were fixed in absolute methanol for 1 min, stained with Giemsa (1:5) for 20 min, air-dried for 4 days, and stored for later analysis of H/L ratios. Two smears per bird were analysed under a light microscope (100×) using immersion oil, and leukocytes were counted as described by Davis (2005). Total erythrocytes per smear were also calculated as the mean number of erythrocytes in six visual fields multiplied by the number of visual fields needed to reach 100 leukocytes. Leukocyte counts were multiplied by 10,000 and divided by total erythrocytes to standardise to leukocytes/10,000 erythrocytes. The mean counts for both smears were calculated to obtain a single value for each bird. The H/L ratio was calculated as heterophils/10,000 erythrocytes divided by lymphocytes/10,000 erythrocytes.

Plasma corticosterone concentrations were determined by radioimmunoassay, as described by Goymann *et al.* (2007). This assay has previously been validated for several species, including seabirds (Huffeldt *et al.* 2020). All samples were assayed in duplicate in one assay. All replicate samples were extracted twice with freshly distilled dichloromethane after overnight equilibration at 4°C of the replicate with 1500 dpm of the respective tritiated steroid to estimate the extraction recovery. The extraction recovery was $82\% \pm 6.2\%$ (mean $\pm$ standard deviation). Standard curves and sample concentrations were calculated using ImmunoFit 3.0 (Beckmann Inc., Fullerton, CA, USA) using a four-parameter logistic-curve fit. The lower detection limit of the corticosterone assay was 3.25 pg/ml, and all samples were above the detection limit. The intra-assay coefficient of variation for corticosterone was 5.0%, which is a measure of the variation among duplicates within an assay, and the intra-extraction coefficient of variation using chicken plasma pool was 9.1%, a measure of the variation of hormone extraction between control samples within assays.

Male Humboldt penguins are typically heavier than females (De La Puente *et al.* 2013). However, to avoid miss-assignment of sex by using only measurements, the sex of the individuals was determined by molecular methods. Total genomic DNA was extracted from blood samples collected on the FTA cards, using a universal and rapid salt extraction protocol (Aljanabi and Martinez 1997). We used two established primers: 2550 F (5'-GTT ACT GAT TCG TCT ACG AGA-3') and 2718 R (5'-ATT GAA ATG ATC CAG TGC TTG-3'), which amplified the chromo-helicase-DNA binding 1 (*CHD1*) gene (Fridolfsson and Ellegren 1999). Polymerase chain reaction (PCR) analyses were performed in 10 µl, using 5 µl of fast mix (SapphireAmp® Fast PCR Master; Takara Bio USA, Inc., Mountain View, CA, USA), 0.5 µl 2550 F (10 µM) and 0.5 µl 2718 R primers (10 µM), 2 µl ddH₂O, 1 µl bovine serum albumin, and 1 µl of the extracted DNA (50–100 µM). The thermal profile was modified to optimise the products with an initial denaturation step of 94°C for 2 min, followed by 38 cycles of denaturation at 92°C for 30s, annealing at 45–50°C for 45s, and elongation at 72°C for 45s, with final elongation for 5 min at 72°C. The PCR products were separated by agarose gel electrophoresis (2%) and sex differences were detected under UV light. Gel separations were run for 40 min. The gels were read by two or three individuals, and unclear gels were repeated until the results were clear. All samples were run twice to confirm the results and all extractions were successful. Males had a single W product (*CHD1W*) and females had two products, W and Z (*CHD1W* (~660 bp) and *CHD1Z* (~750 bp), similar to those found in other penguins (Zhang *et al.* 2013). One band thus indicated a male and two bands indicated a female.

Tracking and diving data were processed in R 4.0.3. Maximum distance from the colony, total trip duration, and total distance travelled during the foraging trip were calculated per individual. For the diving data, an immersion was defined as the period when the depth exceeded 3 m until the bird's return to the surface, similar to the approach used by Raya Rey *et al.* (2013). This threshold may omit shallow foraging dives, but a lower threshold would risk including transiting dives (Sutton *et al.* 2020). The diving depth and duration were calculated for each immersion using the package 'spheniscus' (Lerma 2021). The mean diving depth, dive duration, and diving rate (dives/trip duration) were calculated per individual.

Statistical analyses

Sex differences in plasma corticosterone levels, H/L ratios, and foraging and diving trip parameters were

compared using two-sample *t*-tests. The Shapiro-Wilk test was used to test for normality. Pre-trip plasma corticosterone levels and trip duration were log transformed for data normalisation. Data on graphs are presented as nontransformed values. We tested if the pre-trip plasma corticosterone levels and H/L ratios could explain the foraging and diving trip parameters (maximum distance from the colony, total distance, trip duration, mean and maximum diving depth, and dive duration) using linear regression analyses. Both sexes were pooled together for the analyses. We also tested if foraging and diving trip parameters could explain post-trip corticosterone levels using linear regression, for each foraging and diving trip parameter separately. For linear regressions, the normality assumptions were additionally checked for all the tested variables by plotting the residuals against predicted values, and there were no clear indications that the normality assumptions were violated. The significance level was 0.05 for all tests

Results

Humboldt Penguin females did not show higher H/L ratios or higher plasma corticosterone levels than males as we expected, instead, there were no significant differences in H/L ratios (Figure 1(a)), pre-trip plasma corticosterone levels (Figure 1(b)), and post-trip plasma corticosterone levels (Figure 1(c)) between the sexes. We expected differences between females and males in their foraging behaviour, but there was no evidence of sex-related differences in foraging and diving trip parameters (Figure 1(d–h)), except that males dived on average deeper than females (Figure 1(i)). While testing relationships between plasma corticosterone levels before and after foraging trips, the H/L ratio, and foraging and diving behaviours in this species, we found no significant relationships (Table 1).

Discussion

Contrary to our expectation, our results show no significant differences in plasma corticosterone levels and H/L ratios of females and males. No significant differences in these parameters contrast with our hypothesis of females having higher levels of stress due to a limited resilience (Pichegru and Parsons 2014), and with reports that found sex-specific haematological parameters in other penguin species (Moreno *et al.* 2002; Gallo *et al.* 2019). Although we must remain cautious because our sample size was limited and this might have contributed to a lack of detectable differences, no differences in plasma corticosterone levels and H/L ratios between

sexes agreed with results for other penguin species (*Pygoscelis adeliae*, Angelier *et al.* 2008; *Spheniscus magellanicus*, D'Amico and Bertellotti 2019; Palacios *et al.* 2018). No differences in physiological stress levels suggest that female and male Humboldt penguins have a similar response to stress during chick rearing at Tilgo Island.

We found no differences in the foraging and diving behaviours between female and male Humboldt penguins, except that males dived on average deeper than females. This result is in accordance with the literature, as previous studies have also found no sex-specific foraging behaviours of Humboldt penguins except for diving depth (Taylor *et al.* 2002; Raya Rey *et al.* 2013). In Humboldt penguins, as in other seabird species, differences in diving depth are likely effect of heavier individuals diving deeper than lighter ones (Shealer 2001), avoidance of competition (Raya Rey *et al.* 2013), and/or thermal habitat preference of each sex (Ciancio *et al.* 2016).

We found no significant relationships between plasma corticosterone levels and H/L ratios with foraging and diving parameters as we expected, although our sample size was low. This lack of a significant relationship between plasma corticosterone levels and foraging trip parameters contrasts with the results for Adélie penguins (Angelier *et al.* 2008). Adélie penguins' lower pre-trip corticosterone levels were associated with longer foraging trips, but Adélie penguins' foraging trips are longer than those of Humboldt penguins, and the sample size used in this study was also small ($n = 18$). In accordance with our results, most studies using seabirds have found no relationships between corticosterone levels or H/L ratios and the foraging behaviour of the individuals (Sorenson *et al.* 2017; Dunphy *et al.* 2020). As suggested for other seabird species (Dunphy *et al.* 2020), Humboldt penguins' foraging effort is likely sufficient benefit to avoid increases in their stress levels. However, future work increasing the sample size and/or including experimental manipulation would be necessary to test with precision the role of the foraging effort on maintaining Humboldt penguins' stress levels.

In summary, we found no differences in the stress levels between female and male Humboldt penguins, no sex-related differences between most of the foraging parameters, except for diving depth, and no relation between the stress levels and the foraging parameters. We acknowledge that our sample size is small and caution should be taken while interpreting these results. Nevertheless, this study contributes with baseline information on physiological stress levels and foraging behaviour of Humboldt penguins. Interestingly, H/L ratios in Humboldt penguins were within the levels expected

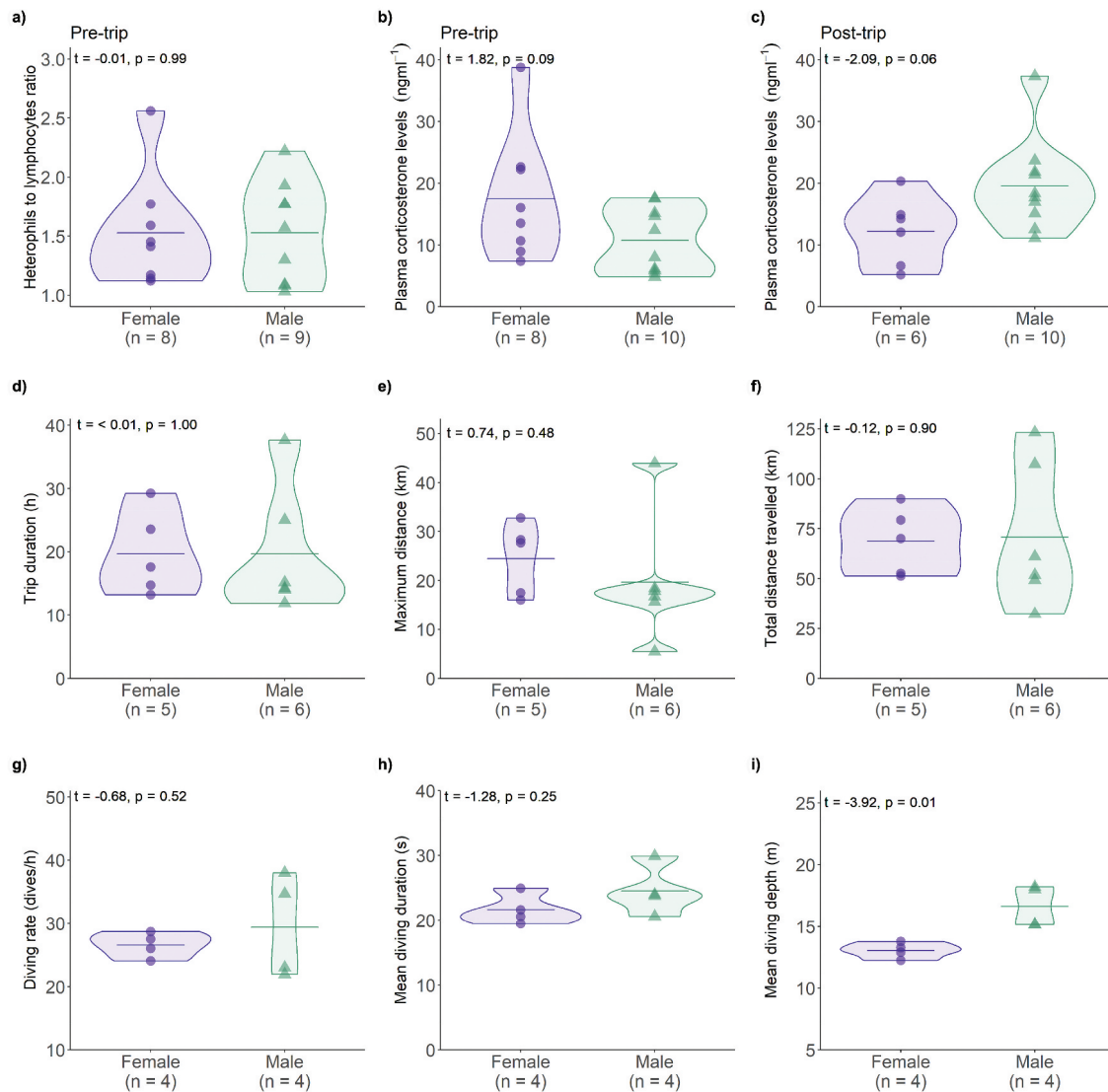


Figure 1. Violin plot of sex differences in a) leukocyte profiles b) pre-trip corticosterone levels, c) post-trip corticosterone levels, d) trip duration, e) maximum distance from the colony, f) total distance travelled, g) diving rate, h) mean diving duration and i) mean diving depth of chick-rearing Humboldt Penguins at Tilgo Island, Northern Chile. Violin plot area shows the distribution of the data at different values. The line shows the mean value. Individual values are shown. Purple circles, females; green triangles, males. Results from two-sample t-tests are presented.

Table 1. Regression analyses of relationships between leukocyte profiles, corticosterone levels, and foraging and diving parameters in Humboldt penguins.

Parameter	H/L ratio			Corticosterone levels (pre-trip)			Corticosterone levels (post-trip)		
	F	P	r ²	F	P	r ²	F	P	r ²
Trip duration	1.53	0.25	0.15	0.25	0.63	0.03	0.02	0.89	0
Maximum distance	0.83	0.39	0.08	0.03	0.86	0.01	0.09	0.77	0.01
Total distance	2.92	0.12	0.25	0.26	0.62	0.03	0.01	0.89	< 0.01
Diving rate	0.35	0.58	0.06	0.14	0.72	0.02	1.12	0.34	0.18
Diving duration	1.33	0.29	0.18	1.62	0.25	0.21	1.03	0.36	0.17
Diving depth	0.05	0.83	0.01	4.65	0.07	0.44	0.97	0.37	0.16

Sexes were pooled together.

for banded penguins (*Spheniscus* spp.) (Moreno *et al.* 2002; D'Amico and Bertellotti 2019; Gallo *et al.* 2019; Minias 2019), but plasma corticosterone levels at Tilgo

Island were higher than those of Humboldt penguins at Punta San Juan (Walker *et al.* 2015). The reasons for differences in plasma corticosterone levels with the

literature are beyond the scope of this study and merit further investigation. In regards to foraging parameters, just a couple of studies have considered sex while examining the foraging behaviour of Humboldt Penguins (e.g. Taylor *et al.* 2002; Raya Rey *et al.* 2013), therefore, our results contribute to the body of literature available for the species. This information might be of particular importance for marine spatial planning and to reduce the risk of differential threats according to depths used by each sex (Taylor *et al.* 2002; Wearmouth and Sims 2008; Crawford *et al.* 2017). Finally, finding no relation between foraging and stress levels would suggest that, at least during the period of the study, each sex was able to regulate their stress levels and foraging effort. This, however, might change at other environmental conditions, breeding stages, colonies, or levels of disturbance. Therefore, comparative studies such as those in Common Diving Petrel (*Pelecanoides urinatrix*) in contrasting colonies (Dunphy *et al.* 2020), and in Magellanic Penguin at contrasting stages of breeding (Raya Rey *et al.* 2012; Colominas-Ciuró *et al.* 2022) are further needed.

Acknowledgements

We thank Wolfgang Goymann and Monika Trappschuh from the Max Planck Institute for Ornithology in Germany for their collaboration with the hormone analysis. Special thanks go to Nicolas Gouin and Paula Plaza for their professional advice regarding the sexing analysis and the Centro de Estudios Avanzados en Zonas Áridas (CEAZA) for allowing us to use their facilities to conduct this analysis. We thank Esther Rickert for the transportation of tracking devices, and Henri Kröling for useful discussions. We thank the reviewers who provided thoughtful recommendations that improved the manuscript.

Disclosure statement

No potential conflict of interest was reported by the authors.

Ethics statement

This study was conducted according to Subsecretaría de Pesca y Acuicultura de Chile permit resolution no. 3367/2017, in line with the applicable international ethical standards for the care and use of wild animals. The procedures were performed by qualified veterinarians under the regulations and supervision of local authorities.

Funding

This study was financed by Fondecyt postdoctorate (grant no. 3170936 to RQ) and the Research and Technology Centre (FTZ), University of Kiel. ML received support from CONACyT-INAPI (grant no. 411876). CV received support

from Fondecyt postdoctorate (grant no. 3160679). RQ received support from Convocatoria Nacional Subvención a la Instalación en la Academia 2021 (grant no. SA77210088-ANID-Chile).

ORCID

Miriam Lerma  <http://orcid.org/0000-0002-7632-9289>
 Nicolas Luna  <http://orcid.org/0000-0002-0338-8333>
 Matías Portflitt-Toro  <http://orcid.org/0000-0001-7850-0628>
 Juan Serratos  <http://orcid.org/0000-0001-9980-919X>
 Guillermo Luna-Jorquera  <http://orcid.org/0000-0003-4274-7025>
 Stefan Garthe  <http://orcid.org/0000-0003-0548-9346>
 Rene Quispe  <http://orcid.org/0000-0002-3390-8627>

References

- Aljanabi, S. M., and Martinez, I. (1997). Universal and rapid salt-extraction of high quality genomic DNA for PCR-based techniques. *Nucleic Acids Research* **25**, 4692–4693. doi:10.1093/nar/25.22.4692
- Angelier, F., Bost, C. A., Giraudeau, M., Bouteloup, G., Dano, S., and Chastel, O. (2008). Corticosterone and foraging behavior in a diving seabird: The Adélie penguins (*Pygoscelis adeliae*). *General and Comparative Endocrinology* **156**, 134–144. doi:10.1016/j.ygcen.2007.12.001
- Angelier, F., Shaffer, S. A., Weimerskirch, H., Trouvé, C., and Chastel, O. (2007). Corticosterone and foraging behaviour in a pelagic seabird. *Physiological and Biochemical Zoology* **80**, 283–292. doi:10.1086/512585
- Ciancio, J. E., Quintana, F., Sala, J. E., and Wilson, R. P. (2016). Cold birds under pressure: Can thermal substitution ease heat loss in diving penguins. *Marine Biology* **163**, 43. doi:10.1007/s00227-016-2813-2
- Colominas-Ciuró, R., Bertellotti, M., D'Amico, V. L., Carabajal, E., Benzal, J., Vidal, V., Motas, M., *et al.* (2022). Sex matters? Association between foraging behaviour, diet, and physiology in Magellanic penguins. *Marine Biology* **169**, 21. doi:10.1007/s00227-021-04003-w
- Crawford, R., Ellenberg, U., Frere, E., Hagen, C., Baird, K., Brewin, P., Crofts, S., *et al.* (2017). Tangled and drowned: A global review of penguin bycatch in fisheries. *Endangered Species Research* **34**, 373–396. doi:10.3354/esr00869
- D'Amico, V. L., and Bertellotti, M. (2019). Physiological toolkit to assess the health impacts on a Magellanic Penguin (*Spheniscus magellanicus*) colony located in a high-use touristic region of Patagonia, Argentina. *Marine Ornithology* **47**, 247–251.
- Dantzer, B., Fletcher, Q. E., Boonstra, R., and Sheriff, M. J. (2014). Measures of physiological stress: A transparent or opaque window into the status, management and conservation of species? *Conservation Physiology* **2**, cou023. doi:10.1093/conphys/cou023
- Davis, A. K. (2005). Effect of handling time and repeated sampling on avian white blood cell counts. *Journal of Field Ornithology* **76**, 334–338. doi:10.1648/0273-8570-76.4.334

- Davis, A. K., Maney, D. L., and Maerz, J. C. (2008). The use of leukocyte profiles to measure stress in vertebrates: A review for ecologists. *Functional Ecology* **22**, 760–772. doi:10.1111/j.1365-2435.2008.01467.x
- De La Puente, S., Bussalleu, A., Cardeña, M., Valdés-Velásquez, A., Majluf, P., and Simeone, A. (2013). Humboldt Penguin (*Spheniscus humboldti*). In 'Penguins: Natural History and Conservation.' (Eds P. Garcia Borboroglu and P. D. Boersma.) pp. 15–287. (University of Washington: Seattle, USA.)
- Dunphy, B. J., Vickers, S. I., Zhang, J., Sagar, R. L., Landers, T. J., Bury, S. J., Hickey, A. J. R., et al. (2020). Seabirds as environmental indicators: Foraging behaviour and ecophysiology of Common diving petrels (*Pelecanoides urinatrix*) reflect local-scale differences in prey availability. *Marine Biology* **167**, 1–12. doi:10.1007/s00227-020-3672-4
- Ellenberg, U., Mattern, T., Seddon, P. J., and Luna-Jorquera, G. (2006). Physiological and reproductive consequences of human disturbance in Humboldt penguins: The need for species-specific visitor management. *Biological Conservation* **133**, 95–106. doi:10.1016/j.biocon.2006.05.019
- Ellenberg, U., Setiawan, A. N., Cree, A., Houston, D. M., and Seddon, P. J. (2007). Elevated hormonal stress response and reduced reproductive output in Yellow-eyed penguins exposed to unregulated tourism. *General and Comparative Endocrinology* **152**, 54–63. doi:10.1016/j.ygcen.2007.02.022
- Fridolfsson, A. A., and Ellegren, H. (1999). A simple and universal method for molecular sexing of non-ratite birds. *Journal of Avian Biology* **30**, 116–121. doi:10.2307/3677252
- Gallo, L., Vanstreels, R. E. T., Cook, R. A., Karesh, W. B., and Uhart, M. (2019). Haematology, plasma biochemistry, and trace element reference values for free-ranging adult Magellanic penguins (*Spheniscus magellanicus*). *Polar Biology* **42**, 733–742. doi:10.1007/s00300-019-02467-7
- Goymann, W., Schwabl, I., Trappschuh, M., and Hau, M. (2007). Use of ethanol for preserving steroid and indoleamine hormones in bird plasma. *General and Comparative Endocrinology* **150**, 191–195. doi:10.1016/j.ygcen.2006.09.014
- Huffeldt, N. P., Merkel, F. R., Jenni-Eiermann, S., Goymann, W., and Helm, B. (2020). Melatonin and corticosterone profiles under polar day in a seabird with sexually opposite activity-rhythms. *General and Comparative Endocrinology* **113296**, doi:10.1016/j.ygcen.2019.113296
- Kitaysky, A. S., Piatt, J. F., Hatch, S. A., Kitaikaia, E. V., Benowitz-Fredericks, Z. M., Shultz, M. T., and Wingfield, J. C. (2010). Food availability and population processes: Severity of nutritional stress during reproduction predicts survival of long-lived seabirds. *Functional Ecology* **24**, 625–637. doi:10.1111/j.1365-2435.2009.01679.x
- Kitaysky, A. S., Piatt, J. F., and Wingfield, J. C. (2007). Stress hormones link food availability and population processes in seabirds. *Marine Ecology Progress Series* **352**, 245–258. doi:10.3354/meps07074
- Landys, M. M., Ramenofsky, M., and Wingfield, J. C. (2006). Actions of glucocorticoids at a seasonal baseline as compared to stress-related levels in the regulation of periodic life processes. *General and Comparative Endocrinology* **148**, 132–149. doi:10.1016/j.ygcen.2006.02.013
- Lerma, M. (2021). Package 'spheniscus' (Version v1.0). doi:10.5281/zenodo.4724480
- Minias, P. (2019). Evolution of heterophil/lymphocyte ratios in response to ecological and life-history traits: A comparative analysis across the avian tree of life. *Journal of Animal Ecology* **88**, 554–565. doi:10.1111/1365-2656.12941
- Moreno, J., Yorio, P., Garcia-Borboroglu, P., Potti, J., and Villar, S. (2002). Health state and reproductive output in Magellanic penguins (*Spheniscus magellanicus*). *Ethology Ecology & Evolution* **14**, 19–28. doi:10.1080/08927014.2002.9522758
- Palacios, M. G., D'Amico, V. L., and Bertellotti, M. (2018). Ecotourism effects on health and immunity of Magellanic penguins at two reproductive colonies with disparate touristic regimes and population trends. *Conservation Physiology* **6**, 1–13. doi:10.1093/conphys
- Petersen, S. L., Ryan, P. G., and Gremillet, D. (2006). Is food availability limiting African Penguins *Spheniscus demersus* at Boulders? A comparison of foraging effort at mainland and island colonies. *Ibis* **148**, 14–26. doi:10.1111/j.1474-919X.2006.00459.x
- Pichegru, L., and Parsons, N. J. (2014). Female-biased mortality in African penguins. *African Journal of Marine Science* **36**, 279–282. doi:10.2989/1814232X.2014.920728
- Quispe, R., Lerma, M., Luna, N., Portflitt-Toro, M., Serratos, J., and Luna-Jorquera, G. (2020). Foraging ranges of Humboldt penguins *Spheniscus humboldti* from Tilgo Island: The critical need for protecting a unique marine habitat. *Marine Ornithology* **48**, 205–208.
- Raya Rey, A., Pütz, K., Scioscia, G., Lüthi, B., and Schiavini, A. (2012). Sexual differences in the foraging behaviour of Magellanic Penguins related to stage of breeding. *Emu - Austral Ornithology* **112**, 90–96. doi:10.1071/MU11065
- Raya Rey, A., Pütz, K., Simeone, A., Hiriart-Bertrand, L., Reyes-Arriagada, R., Riquelme, V., and Lüthi, B. (2013). Comparative foraging behaviour of sympatric Humboldt and Magellanic Penguins reveals species-specific and sex-specific strategies. *Emu - Austral Ornithology* **113**(2), 145–153. doi:10.1071/MU12040
- Shealer, D. A. (2001). Foraging behaviour and food of seabirds. In 'Biology of Marine Birds.' (Eds E. A. Schreiber and J. Burger.) pp. 137–170. (CRC Press : Boca Raton.)
- Sorenson, G. H., Dey, C. J., Madliger, C. L., and Love, O. P. (2017). Effectiveness of baseline corticosterone as a monitoring tool for fitness: A meta-analysis in seabirds. *Oecologia* **183**, 353–365. doi:10.1007/s00442-016-3774-3
- Sutton, G., Pichegru, L., Botha, J. A., Kouzani, A. Z., Adams, S., Bost, C. A., and Arnould, J. P. Y. (2020). Multi-predator assemblages, dive type, bathymetry and sex influence foraging success and efficiency in African penguins. *PeerJ* **8**, e9380. doi:10.7717/peerj.9380
- Taylor, S. S., Leonard, M. L., Boness, D. J., and Majluf, P. (2002). Foraging by Humboldt penguins (*Spheniscus humboldti*) during the chick-rearing period: General patterns, sex differences, and recommendations to reduce incidental catches in fishing nets. *Canadian Journal of Zoology* **80**, 700–707. doi:10.1139/z02-046
- Walker, B. G., Pee Boersma, P., and Wingfield, J. C. (2015). The glucocorticoid stress response in Magellanic penguins (*Spheniscus magellanicus*): Comparing within and between breeding seasons, by age and colony, after fighting, and with other penguin species. *Canadian Journal of Zoology* **93**, 123–131. doi:10.1139/cjz-2014-0216

- Wearmouth, V. J., and Sims, D. W. (2008). Sexual segregation in marine fish, reptiles, birds and mammals: Behaviour patterns, mechanisms and conservation implications. *Advances in Marine Biology* **54**, 107–170. doi:[10.1016/S0065-2881\(08\)00002-3](https://doi.org/10.1016/S0065-2881(08)00002-3)
- Wilson, R. P., Pütz, K., Peters, G., Culik, B., Scolaro, J. A., Charrassin, J. B., and Ropert-Coudert, Y. (1997). Long-term attachments of transmitting and recording devices to penguins and other seabirds. *Wildlife Society Bulletin* **25**, 101–106.
- Yamamoto, T., Yoda, K., Blanco, G. S., and Quintana, F. (2019). Female-biased stranding in Magellanic penguins. *Current Biology* **29**, R12–R13. doi:[10.1016/j.cub.2018.11.023](https://doi.org/10.1016/j.cub.2018.11.023)
- Zhang, P., Han, J., Liu, Q., Zhang, J., and Zhang, X. (2013). Sex identification of four penguin species using locus-specific PCR. *Zoo Biology* **32**, 257–261. doi:[10.1002/zoo.21005](https://doi.org/10.1002/zoo.21005)